Do Now · retrieval

Thinking logically: decisions and program flow

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Why must a booking be saved before confirmation?

Q2

What is true AND false?

Q3

What does an if condition control?

Q4

What is false OR true?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Locate the decision — A discount needs BOTH an eligible membership and a basket of at least £50.

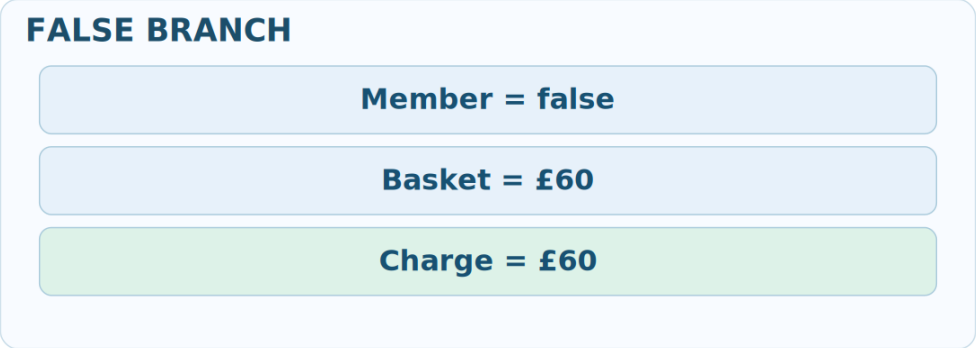
Core idea 2

Follow the true branch — For member=true and basket=60, the condition is true and 10% discount produces £54.

Core idea 3

Follow the false branch — For member=false with basket=60, the AND expression is false, so the full £60 is charged.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

condition — Expression evaluated as true or false.

branch — Path selected by a decision.

AND — True only if both operands are true.

OR — True if either or both operands are true.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

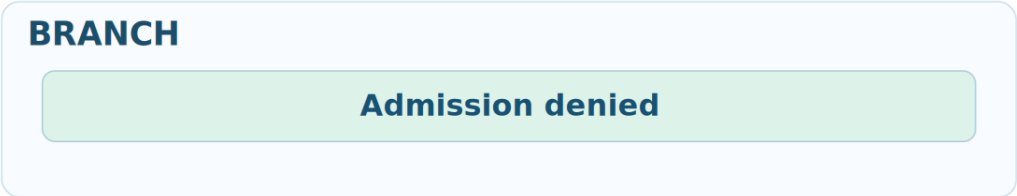
Problem

A cinema permits admission for age≥15 AND a valid ticket. Trace age=16 with no ticket.

Follow the worked method

1. Predict first — The age condition is true, but the ticket condition is false.
2. Model the reasoning — AND requires both true; the result is false, so the deny branch executes.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first

The score condition is false because 72<80.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Give A AND B for (false,false), (false,true), (true,false), (true,true).

Question 2

Access is allowed if Year13 AND (HasCode OR TeacherApproved). Evaluate Year13=true, HasCode=false, TeacherApproved=true; then Year13=false, HasCode=true, TeacherApproved=true.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

A developer changes the discount condition `member && basket>=50` to `member || basket>=50`. Give two wrong outcomes and correct the expression.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Replacing AND with OR in a two-part eligibility rule can permit extra cases.

Use key vocabulary from memory

1. Explain condition in your own words.

2. Explain branch in your own words.

3. Explain AND in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.